

Saddle-node bifurcation of first-passage-time densities induced by a directed shortcut

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(Dated: July 2, 2026)

First-passage-time (FPT) densities in shortcut-assisted search can be bimodal: a fast “direct” arrival and a slow “indirect” one. We show that this bimodality is the observable-level signature of a *saddle-node bifurcation* that can be solved exactly. Modeling a directed shortcut to an absorbing target as a rank-one killing defect on a lazy lattice—whose diffusive limit is Brownian motion on an interval with an interior δ -sink of strength b at fractional position θ —we find that the FPT density Φ carries a short-time shortcut-capture mode together with a finite-time diffusive-return peak, and that this second, interior peak is created and annihilated through a fold of a *signed* spectral mixture governed by the compact criterion $S_1 = S_2 = 0$, where $S_n(\tau) = \sum_j G_j \mu_j^n e^{-\mu_j \tau}$ and $\Phi' = -S_1$, $\Phi'' = S_2$. Numerical continuation gives a single connected two-mode window $0 < b < b_c(\theta; \xi)$ (source-started, $\xi = \theta$), with a computable boundary $b_c(\theta)$ that is symmetric about $\theta = \frac{1}{2}$, minimal near $\theta \simeq 0.381$, and diverges as $b_c \sim 0.789/\min(\theta, 1 - \theta)$ near the endpoints; the fold obeys the generic saddle-node scaling $\tau_+ - \tau_- \propto (b_c - b)^{1/2}$, $\Delta\Phi \propto (b_c - b)^{3/2}$, requires at least three modes, and persists when the walker is released away from the gate, where both features become genuine interior peaks. The mechanism is structurally stable rather than lattice-specific: it follows from a low-rank killing defect, it admits a rank-two extension in which a cusp organizes a triple-peak region, and finite two-dimensional lattice simulations show the same fold morphology, and an independent continuum Brownian-dynamics simulation—no lattice, no spectral input—reproduces both the exact density and the fold; a direct Monte-Carlo and an exact channel decomposition identify the early feature as shortcut capture and the late peak as diffusive return. The threshold b_c is a time-domain morphology fold, *not* a spectral exceptional point. The result recasts a known phenomenology as a predictive, exactly solvable bifurcation with a measurable threshold and generic scaling, testable in feedback-controlled colloidal search and engineered transport networks.

I. INTRODUCTION

The first-passage time (FPT) to a target is a central observable of stochastic transport, search and reaction kinetics [1–3]. Its full distribution $f(t)$, rather than the mean first-passage time (MFPT) alone, controls the timing statistics of diffusion-limited reactions, of intracellular delivery, and of search under deadlines, and it is increasingly accessible experimentally through single-particle tracking and feedback-controlled colloidal platforms [9, 10].

A recurring and striking feature of $f(t)$ is *multimodality*. When a searcher has access to a “shortcut”—a relocation channel, a long-range rewired edge, or an active-transport route—the FPT density can become bimodal: fast trajectories that use the shortcut form an early peak, while slow trajectories that explore the domain form a later peak, provided the two time scales are well separated [6]. Random walks on small-world and shortcut-rich networks likewise exhibit shortcut-induced temporal structure in their first-passage statistics [8]. Multimodal capture-time densities are further known for interior partially absorbing traps when several traps are strategically arranged [9, 10], and exact spatiotemporal solutions of confined lattice random walks with defects are by now a mature program [4, 5].

These works analyze direct/indirect first-passage structure and partial-absorption kinetics [6, 10]; what has been missing is an exactly solvable model whose FPT-density *extrema* undergo a certified saddle-node fold at a computable shortcut-strength threshold. That is the gap we fill. The distinction matters: the appearance and disappearance of a finite-time arrival peak is a qualitative, controllable property of the distribution that is invisible to the MFPT and to the spectral gap, and is, we argue, a genuine catastrophe of the observable.

Here we supply exactly that classification for a minimal directed shortcut, modeled as a rank-one killing defect on an absorbing lattice. Our contributions are:

1. An *exact solution* (Sec. II): the FPT density is a signed spectral mixture whose amplitudes we derive in closed form from the defect resolvent, both on the finite lattice (Montroll determinant, App. A) and in the diffusive limit (App. B).
2. A *saddle-node fold criterion* (Sec. III): the FPT density carries a short-time shortcut-capture mode and a finite-time diffusive-return peak; the birth and death of the latter is a fold governed by the compact local criterion $S_1 = S_2 = 0$ in signed spectral moments, with $\Phi' = -S_1$, $\Phi'' = S_2$. For the source-started convention ($\xi = \theta$), numerical continuation gives a single connected two-mode window $0 < b < b_c(\theta; \xi)$.
3. A *computable phase boundary* $b_c(\theta)$: symmetric

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about $\theta = \frac{1}{2}$, minimized near $\theta \simeq 0.381$, with the endpoint law $b_c \sim 0.789/\min(\theta, 1 - \theta)$; a *minimal-mode theorem* (at least three modes are required); and the generic saddle-node scaling $\tau_+ - \tau_- \propto (b_c - b)^{1/2}$, $\Delta\Phi \propto (b_c - b)^{3/2}$.

4. A *structural-stability/universality* argument (Sec. IV): for reversible (symmetrizable) low-rank killing the fold is model-independent, admits a rank-two extension in which a cusp organizes a triple-peak region, reproduces on finite two-dimensional lattices, and is realized in an independent continuum Brownian-dynamics simulation (no lattice, no spectral input).
5. A *physical embedding* (Sec. V): a direct Monte-Carlo and an exact channel decomposition identify the early feature as shortcut capture and the late peak as diffusive return; the fold persists for starts away from the gate; and the threshold b_c is a time-domain morphology fold, *not* a spectral exceptional point.

We are careful to separate what is new from what is not. Shortcut-induced bimodality and interior-trap multimodality are known [6, 9, 10]; the lattice Green-function/defect method is Giuggioli's [4, 5]; and Ref. [7] concerns bimodality of a splitting probability $P(\omega)$, not of $f(t)$. Our contribution is the exactly solvable *bifurcation-theoretic* account: the fold criterion, the predictive threshold $b_c(\theta)$, and the universality of the mechanism.

II. MODEL AND EXACT SOLUTION

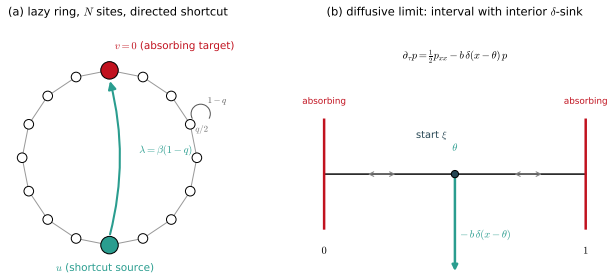


FIG. 1. Model. (a) Lazy random walk on a ring of N sites (stay $1 - q$, hop $q/2$) with an absorbing target $v = 0$ and a directed shortcut $u \rightarrow v$ of rate $\lambda = \beta(1 - q)$; deleting v leaves a rank-one killing defect at u . (b) Diffusive limit: Brownian motion on $[0, 1]$ with absorbing ends and an interior δ -sink of strength b at θ , started at ξ .

A. Lazy ring with a directed shortcut

Consider a lazy random walk on a ring of N sites [Fig. 1(a)]: at each step the walker stays with probability $1 - q$ and hops to each neighbor with probability $q/2$. Site $v = 0$ is an absorbing target. A directed shortcut at a source site u diverts probability $\lambda = \beta(1 - q)$ from the u self-loop onto the directed edge $u \rightarrow v$; upon taking it the walker is absorbed. Removing the absorbed state leaves a transient (sub-stochastic) generator on sites $1, \dots, N - 1$ that is a Dirichlet path with a rank-one diagonal killing defect of strength $\lambda = \beta(1 - q)$ at u (admissibly $0 \leq \lambda \leq 1 - q$, as it is diverted from the self-loop probability),

$$Q_\lambda = Q_0 - \lambda |u\rangle\langle u|, \quad (1)$$

so probability leaks irreversibly from u into the target; the rescaled continuum sink strength $b = \lambda N/q$ is introduced below. Although the shortcut is directed, the transient block remains symmetric—the directedness enters only through the (one-way) absorption vector, not the survival dynamics—so its spectrum is real. The eliminated survival operator is therefore self-adjoint: the process is open (sub-Markovian, probability-losing), but the threshold is a time-domain morphology change, not a non-Hermitian spectral (exceptional-point) effect (Sec. V C).

B. Exact generating function and Montroll determinant

The FPT generating function is rational with an exact Montroll-type determinant. Applying the matrix-determinant lemma to the rank-one defect (1) and evaluating the resulting path-resolvent element in closed form (App. A) gives, with U_m the Chebyshev polynomials of the second kind, $y = 1 + (z^{-1} - 1)/q$ the spectral variable, and $a = q/\lambda$ the dimensionless inverse shortcut strength, the characteristic determinant for a source at u ,

$$D_u(y) = a U_{N-1}(y) + 2 U_{u-1}(y) U_{N-u-1}(y). \quad (2)$$

The FPT probability is then a finite sum of geometric modes,

$$F(t) = \sum_j B_j s_j^{t-1}, \quad s_j = 1 - q + q y_j, \quad (3)$$

where y_j are the roots $D_u(y_j) = 0$ and the residues are $B_j = q \mathcal{N}_{r,u}(y_j)/D'_u(y_j)$ with r the start site and $\mathcal{N}_{r,u}$ the piecewise numerator of App. A. The antipodal case $u = N/2 \equiv L$ reduces (2) to $D \propto a T_L + U_{L-1}$ (with T_L the Chebyshev polynomial of the first kind), recovering the reflection-symmetric problem as a special case of the general rank-one theory. Equivalently, with the transient transition matrix $P_\lambda = P_0 - \lambda |u\rangle\langle u|$, the total absorption current has the compact form

$$\tilde{F}_r(z) = z \left[\lambda \langle u| + \frac{q}{2} (\langle 1| + \langle N-1|) \right] (I - z P_\lambda)^{-1} |r\rangle, \quad (4)$$

whose two bracketed terms are the shortcut-capture and diffusive-absorption channels resolved in Sec. IV E. In the diffusive scaling the finite mass maps to a continuum density, $F(t) \simeq (q/N^2) \Phi(qt/N^2)$ with $\tau = qt/N^2$.

C. Diffusive limit

Writing $\theta = u/N$ for the fractional shortcut position, $b = \beta(1-q)N/q$ for the rescaled sink strength, $\tau = qt/N^2$ for diffusive time, and expanding $s = 1 - q + q \cos(2w/N)$, the walk converges to Brownian motion on $[0, 1]$ with absorbing ends and an interior δ -sink [Fig. 1(b)],

$$\partial_\tau p = \frac{1}{2} p_{xx} - b \delta(x - \theta) p, \quad p(0) = p(1) = 0. \quad (5)$$

Its spectrum is fixed by the transcendental determinant

$$D_\theta(k; b) = k \sin k + 2b \sin(k\theta) \sin(k(1 - \theta)) = 0, \quad (6)$$

with $k = 2w$ and eigenvalues $\mu_j = k_j^2/2 = 2w_j^2$; for $\theta = \frac{1}{2}$ this collapses to $\tan w = -2w/b$. This interval δ -sink (point-interaction) eigenvalue condition is the standard point-sink spectrum [9, 10]; our exact-solution novelty is not this spectrum but the finite- N directed-shortcut Montroll determinant D_u [Eq. (2)] and the bifurcation classification developed below. The FPT density from a start at $\xi = r/N$ is the signed spectral mixture

$$\Phi_{\xi, \theta}(\tau; b) = \sum_j G_j e^{-\mu_j \tau}, \quad (7)$$

whose amplitudes we derive from the defect resolvent (App. B) as

$$G_{\xi, \theta} = 2w^2 \frac{\phi_{k, \theta}(\xi) I(k, \theta)}{J(k, \theta)}, \quad k = 2w, \quad (8)$$

where

$$\phi_{k, \theta}(x) = \begin{cases} \sin(k(1 - \theta)) \sin(kx), & x \leq \theta, \\ \sin(k\theta) \sin(k(1 - x)), & x \geq \theta, \end{cases} \quad (9)$$

and $I = \int_0^1 \phi dx$, $J = \int_0^1 \phi^2 dx$ have the closed forms of App. B. Modes with $\sin(n\pi\theta) = 0$ do not couple to the sink and contribute the free Dirichlet term $G_n^{\text{node}} = n\pi[1 - (-1)^n] \sin(n\pi\xi)$ (even n vanish). We have verified (8), the crucial normalization identity $J = \sin(k) D_k/(4b)$, and the full master curve (affected plus node modes) against exact finite- N residues to a relative error $\lesssim 6 \times 10^{-5}$ (per-mode $\sim 2 \times 10^{-4}$) with clean $O(N^{-2})$ convergence (App. B, App. E).

III. SADDLE-NODE FOLD CRITERION AND PHASE BOUNDARY

A. Signed spectral moments and the fold criterion

Define the signed spectral moments

$$S_n(\tau; b) = \sum_j G_j \mu_j^n e^{-\mu_j \tau}, \quad (10)$$

so that $\Phi' = -S_1$ and $\Phi'' = S_2$ [differentiate (7)]. We are precise about what folds. Under the source-started convention $\xi = \theta$ the density carries a short-time shortcut-capture feature—a boundary/initial-time mode whose weight vanishes as $b \rightarrow 0$ —together with *interior* finite-time extrema; an interior extremum is a zero of S_1 . The finite-time (diffusive-return) peak forms a valley-peak pair that is born or annihilated when the two interior extrema merge—a *saddle-node* fold of the density's critical points,

$$\boxed{S_1(\tau_c, b_c) = 0, \quad S_2(\tau_c, b_c) = 0}, \quad (11)$$

with nondegeneracy $S_3(\tau_c, b_c) \neq 0$ and $\partial_b S_1(\tau_c, b_c) \neq 0$. No positivity is assumed—the G_j are genuinely signed—so (11) is the correct fold criterion for a signed exponential mixture: the coincidence of a root and a critical point of the generalized (signed) exponential sum S_1 . The amplitudes G_j , and hence the boundary, depend on the start, so $b_c = b_c(\theta; \xi)$; unless stated otherwise we report $\xi = \theta$. The local fold (11) is exact; the global statement is *numerical*: for $\xi = \theta$, numerical continuation of the interior extrema gives a single connected two-mode window $0 < b < b_c(\theta)$ with no re-entrant interval over the scanned range (App. E). We use theorem language only for the local fold and the exact finite- N algebra.

B. The existence boundary $b_c(\theta)$

Eliminating τ_c between the two equations in (11) yields a scalar boundary $b_c(\theta)$ [Fig. 2(a)]. It is symmetric, $b_c(\theta) = b_c(1 - \theta)$, has a minimum at $\theta \simeq 0.381$ ($b_c \simeq 2.16$), and obeys the endpoint law

$$b_c(\theta) \sim \frac{B^*}{d}, \quad \tau_c \sim c_* d^2, \quad d = \min(\theta, 1 - \theta), \quad (12)$$

with an independently resolved constant $B^* \approx 0.78903$ (and $c_* \approx 0.158$) from the half-line boundary-layer problem of App. C: bisection on the fold gives $b_c\theta = 0.789026$ across $\theta \in [0.20, 0.27]$ (six digits, no reference table); a closed form for B^* remains open. In the antipodal case

$$b_c(\frac{1}{2}) = 3.07643, \quad \tau_c = 0.038363, \quad (13)$$

independently certified to 10^{-6} , with the near-antipodal expansion $b_c(\frac{1}{2} + \epsilon) \approx 3.0764 - 133\epsilon^2$. The two limits are physically transparent: as $b \rightarrow 0$ the capture mode's weight vanishes and the density reverts to the single-peaked Dirichlet FPT, while as $b \rightarrow \infty$ the δ -sink becomes an absorbing cut and, for $\xi = \theta$, the density collapses toward $\tau \rightarrow 0$ —so the interior second peak exists only in the finite window $0 < b < b_c$.

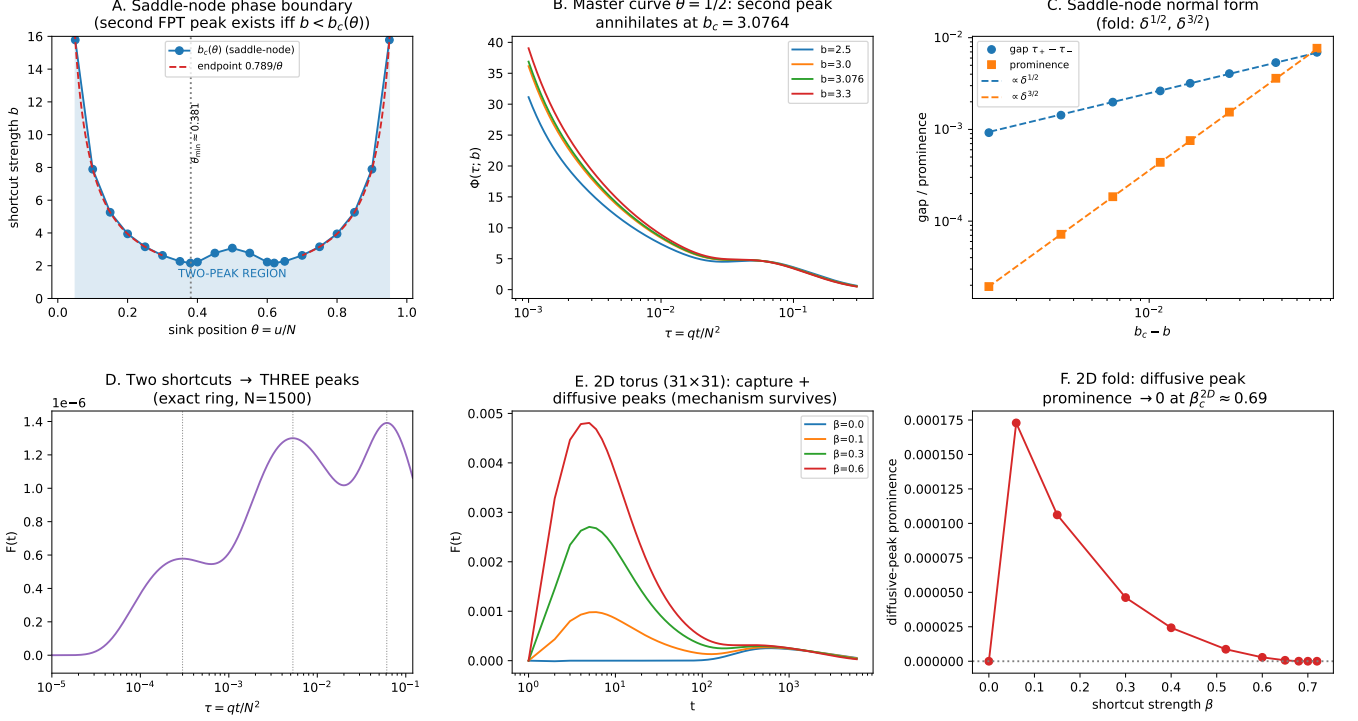


FIG. 2. Results. (a) Phase boundary $b_c(\theta)$ ($\xi = \theta$): the interior second peak exists for $0 < b < b_c(\theta)$; symmetric, minimum at $\theta \simeq 0.381$; solid points are numerically located, and the small- θ branch follows the dashed endpoint asymptote $0.789/\min(\theta, 1 - \theta)$. (b) Master curve $\Phi(\tau; b)$ at $\theta = \frac{1}{2}$ as b crosses $b_c \simeq 3.076$. (c) Normal-form scaling: gap $\propto (b_c - b)^{1/2}$ and prominence $\propto (b_c - b)^{3/2}$ (fitted exponents 0.50, 1.49). (d) Two shortcuts produce a third peak (exact ring, $N = 1500$). (e) 2D torus (31×31): capture and diffusive peaks. (f) 2D fold: the diffusive-peak prominence vanishes near $\beta_c^{2D} \approx 0.69$ ($L = 31$).

C. Normal-form scaling

Near the fold the two merging extrema split with the generic saddle-node exponents

$$\tau_+ - \tau_- \propto (b_c - b)^{1/2}, \quad \Phi(\tau_+) - \Phi(\tau_-) \propto (b_c - b)^{3/2}, \quad (14)$$

which we confirm numerically [Fig. 2(c)], the fitted exponents being 0.50 and 1.49; in the antipodal case the prefactors are $\tau_+ - \tau_- \simeq 0.025 (b_c - b)^{1/2}$ and prominence $\simeq 0.36 (b_c - b)^{3/2}$. The exponents certify the transition as a fold rather than a smooth spectral crossover, and—being the generic saddle-node exponents—constitute a system-independent experimental signature (Sec. V).

D. Minimal mode count

A two-mode fold is impossible, even with signs. Writing $A_j = G_j \mu_j e^{-\mu_j \tau}$, $S_1 = 0$ forces $A_2 = -A_1$, whence $S_2 = (\mu_1 - \mu_2) A_1 = 0$ gives $A_1 = A_2 = 0$, a contradiction. Thus *at least three* modes are required, and the fold ratio $A_1 : A_2 : A_3 = (\mu_3 - \mu_2) : (\mu_1 - \mu_3) : (\mu_2 - \mu_1)$ forces *alternating signs* among the three participating contributions.

The full fold is realized by a many-mode signed cancellation; at the antipode this coincides with an alternating affected-mode pattern ($G_1 > 0, G_2 < 0, G_3 > 0, \dots$), while away from the antipode the modal signs need not be strictly alternating in eigenvalue order (App. B). Retaining K affected modes in the antipodal case gives $b_c^{(2)}$ (none), $b_c^{(3)} = 3.046$, $b_c^{(4)}$ (none), $b_c^{(5)} = 3.0764$ —the even- K gaps reflecting the odd-mode parity of the alternating structure—converging to (13).

IV. UNIVERSALITY

A. Model-independence

The fold (11) is not special to the ring. For any absorbing Markov generator Q_0 , a rank- m killing defect $Q_B = Q_0 - UBU^\top$ gives, by the Woodbury identity,

$$\det(sI - Q_B) = \det(sI - Q_0) \det[I_m + BW(s)], \quad (15)$$

with $W_{ab}(s) = \langle u_a | (sI - Q_0)^{-1} | u_b \rangle$. For *reversible* (symmetrizable) transient dynamics with diagonal low-rank killing—as for the symmetric ring block here— Q_B has a

real spectrum and the FPT density is again a real signed spectral mixture, whose second-peak birth/death is the fold $S_1 = S_2 = 0$. (For non-reversible directed transient dynamics the determinant identity still holds, but the real modal expansion, and hence the fold classification, requires additional hypotheses.) The Woodbury identity supplies only the finite-rank secular determinant; the universality is the *structural stability* of a generic extrema-pair fold (a codimension-one catastrophe of the scalar density [11]), not a consequence of the identity alone. Accordingly the double-peak region is an *open* set (structurally stable by the implicit-function theorem), and the honest statement is not “every shortcut creates a peak”—false in general—but “whenever a shortcut-induced peak is born or annihilated generically, it does so through a fold of the signed spectral mixture, independent of the underlying geometry.”

B. Multiple shortcuts and the cusp

For two shortcuts at $\theta_1 = \alpha < \theta_2 = \eta$ (positions; β is reserved for the shortcut rate) with strengths b_1, b_2 , the secular determinant is (App. D)

$$\begin{aligned} D_{12}(k) = & k \sin k + 2b_1 \sin(k\alpha) \sin(k(1-\alpha)) \\ & + 2b_2 \sin(k\eta) \sin(k(1-\eta)) \\ & + \frac{4b_1 b_2}{k} \sin(k\alpha) \sin(k(\eta-\alpha)) \sin(k(1-\eta)), \end{aligned} \quad (16)$$

the last term being a genuine two-defect interaction that vanishes as the sources coalesce (giving the additive rule $b = b_1 + b_2$). Two shortcuts can produce a *third* FPT peak: at $(\alpha, \eta, b_1, b_2) = (0.38, 0.48, 1.35, 0.14)$, $\xi = 0.463$, the exact ring ($N = 1500$) shows three peaks at $\tau \simeq 3.0 \times 10^{-4}, 5.3 \times 10^{-3}, 6.1 \times 10^{-2}$ [Fig. 2(d)], corresponding to capture via the near shortcut, capture via the far shortcut, and diffusive arrival. The triple-peak wedge in (b_1, b_2) is organized by a *cusp* $S_1 = S_2 = S_3 = 0$; we locate it at

$$(\alpha, \eta, b_1, b_2, \tau_c) = (0.38, 0.489580512568, 1.35, 1.08389, 0.0012366), \quad (17)$$

with $S_4 \neq 0$ and unfolding Jacobian $\det \partial(S_1, S_2)/\partial(b_1, b_2) = -6.9 \times 10^7 \neq 0$, a genuine codimension-two cusp (App. D). The triple-peak example and the cusp point are *distinct control values in the same rank-two family*—the example demonstrates a three-peak region, while the cusp calculation locates a nearby codimension-two organizing point in the (b_1, b_2) unfolding at fixed geometry (they differ in b_2 , 0.14 vs 1.08, not merely in η). Thus m shortcuts turn the single fold into fold hypersurfaces in b -space, with open regions of two and three peaks organized by a cusp [11]; the secular determinants and residues remain finite-rank and explicitly computable, though the catastrophe locations are found numerically.

C. Two dimensions

On an $L \times L$ torus with an absorbing target and a directed shortcut, the same algebra holds [Eq. (15) with the 2D lattice resolvent]. Numerically (at $L = 31$), a single diffusive peak at $\beta = 0$ acquires a shortcut-induced capture peak; the two coexist over a window, and the second (diffusive) interior maximum then annihilates by extremum merging near $\beta_c^{2D} \approx 0.69$, its prominence falling to zero between $\beta = 0.68$ and 0.70 [Fig. 2(e,f)]—the same fold morphology as in one dimension. One caveat is essential: a single-site sink is marginal in 2D (its self-Green function is logarithmically divergent), so the *bare* threshold is expected to drift as $b_c(L) \sim 1/\log L$; the large- L continuum point-sink limit therefore requires logarithmic renormalization, and we leave a size-independent 2D scaling test (the analogue of (14)) to future work. We thus present the 2D result as finite-lattice evidence that the mechanism is not one-dimensional, not as a proof of 2D universality.

D. Robustness

The fold persists under any perturbation that preserves the nondegeneracy $S_3 \neq 0$, $\partial_b S_1 \neq 0$: to first order $\delta b_c = -(\partial_\varepsilon S_1 / \partial_b S_1) \delta \varepsilon$ for a perturbation ε . Concretely it survives weak hop-rate disorder (which can even help, by lifting accidental dark modes), small shifts of the source or start (Sec. IV E), and the laziness q , which is already scaled out of the reduced threshold and reduced fold time (the raw step time obeys $t_c \sim N^2 \tau_c / q$). It is destroyed by placing the source on a symmetry node (dark modes $G_j = 0$), by source coalescence (rank drop), or by moving the source adjacent to the absorbing boundary. Recall that at $\xi = \theta$ the shortcut-capture feature is a boundary/initial-time mode, so the interior fold concerns the finite-time diffusive-return peak; moving the start off the gate turns the capture feature into an interior peak in its own right (Fig. 5).

E. Direct simulation, channel resolution, and start dependence

Four checks confirm that the fold is a physical mechanism rather than an artifact of the spectral representation. (i) *Direct simulation*. A direct Monte-Carlo of the lazy ring, using no spectral input, reproduces the exact FPT density across four decades in τ [Fig. 3(a)], with the simulated splitting probability $\pi_{sc} = 0.5647 \pm 0.0008$ (4×10^5 walkers) agreeing with the exact 0.5652. (ii) *Channel resolution*. The absorption current splits exactly into a shortcut entry (λ at u) and diffusive entries ($q/2$ at the neighbors of v), $\Phi = f_{sc} + f_{diff}$ [Fig. 3(b)]: f_{sc} is monotone and carries the short-time mass, while the late finite-time bump receives comparable contributions from both channels (diffusive return and delayed

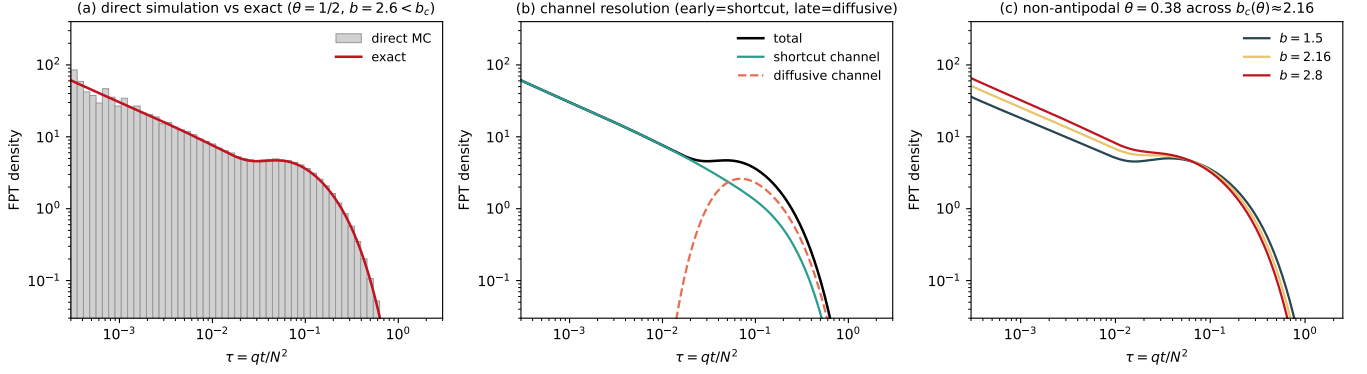


FIG. 3. The fold is a physical mechanism, not an artifact of the exact solution. (a) A direct Monte-Carlo simulation of the lazy ring (grey; 4×10^5 walkers, no spectral input) reproduces the exact FPT density (red) across four decades in τ ; the simulated splitting probability $\pi_{sc} = 0.5647 \pm 0.0008$ agrees with the exact $\pi_{sc} = 0.5652$. (b) The absorption vector splits exactly into a shortcut entry (λ at u) and diffusive entries ($q/2$ at the neighbors of v), $\Phi = f_{sc} + f_{diff}$: f_{sc} is monotone and carries the short-time mass, while the late finite-time bump receives comparable contributions from both channels (diffusive return and delayed capture). (c) The same interior fold occurs away from the antipode, at $\theta = 0.38$: $b_c(0.38)$ lies between the $b = 2.16$ (peak present) and $b = 2.8$ (absent) curves. ($N = 100$, $q = 2/3$, gate $u = N/2$ in (a,b) or $u = 0.38N$ in (c), start $\xi = \theta$.)

capture). The fold thus annihilates the second arrival-time *scale*, not a single pure channel. (iii) *Non-antipodal robustness*. The interior fold occurs off the reflection axis [Fig. 3(c)]: at $\theta = 0.38$, $b_c(0.38)$ lies between the $b = 2.16$ and $b = 2.8$ curves. (iv) *Start dependence*. Releasing the walker away from the gate ($\xi \neq \theta$) removes the boundary/initial capture mode: both features become genuine *interior* peaks that annihilate through the fold [Fig. 5(a)], and the threshold $b_c(\theta; \xi)$ varies smoothly and symmetrically about $\xi = \theta$ [Fig. 5(b)], so the saddle-node is not an artifact of releasing exactly at the gate. (Far from the gate the morphology changes and the simple two-mode window is lost.)

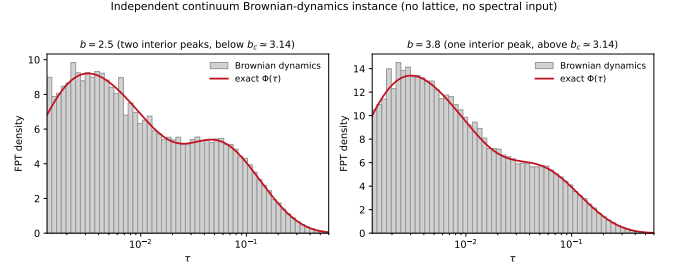


FIG. 4. Independent instance. A continuum Brownian-dynamics simulation of the delivery-gate protocol (6×10^4 walkers, Euler-Maruyama, gate width $\ell = 0.02$, $\theta = \frac{1}{2}$, start $\xi = 0.44$; no lattice, no spectral input) reproduces the exact δ -sink density $\Phi(\tau)$ (red) and the fold: two interior peaks below $b_c \simeq 3.14$ (left) and one above (right).

V. PHYSICAL REALIZATION AND NON-HERMITIAN CHARACTER

F. An independent continuum instance

Finally, the model-independence is realized directly, in a system that is *not* the lazy ring. A continuum Brownian-dynamics simulation of the delivery-gate protocol (Sec. VB)—Brownian motion on $[0, 1]$ with absorbing ends and killing at rate b/ℓ inside a narrow gate of width ℓ at θ , integrated by Euler-Maruyama with *no lattice and no spectral input*—reproduces both the exact δ -sink density and the fold: released off the gate ($\xi = 0.44$), it shows two interior peaks below b_c and one above [Fig. 4], matching the exact continuum $\Phi(\tau)$ to histogram precision. This converts the Woodbury model-independence from an algebraic identity into a demonstrated second instance, and confirms that the fold is not a lattice artifact.

A. Embedding

Physically the shortcut is a localized target-delivery channel. From $p'(t) = Q_0 p - b|u\rangle\langle u|p$ the absorption current is $f(t) = f_0(t) + b p_u(t)$: while the searcher is at u it is irreversibly delivered to the target at rate b . In the survival sector this is a local killing term $-b|u\rangle\langle u|$, not a transport edge, and a finite gate of physical width ℓ with local trigger rate κ coarse-grains to a δ -sink. Restoring dimensions with domain length L , diffusion constant D , and $t = (L^2/D)\tau$, the dimensionless sink strength is

$$b \sim \frac{\kappa \ell L}{D} \quad (\ell \ll L), \quad (18)$$

so a laboratory quadruple (κ, ℓ, L, D) maps to a definite b ; the smallest threshold $b_c(0.38) \simeq 2.16$ corresponds

to a trigger rate $\kappa \simeq 2.16 D/(\ell L)$ at a gate placed at $x \simeq 0.38 L$. The sharp physical statement is that a single localized delivery channel creates an *additional arrival-time scale*—a second interior peak in the full FPT density—above a predictable threshold.

B. Realizations

We rank candidate realizations by cleanliness. (i) *Feedback-controlled colloidal or active-particle search* is the strongest: a narrow gate at $x = \theta L$ (optical tweezers, a microfluidic pulse, or active-particle feedback) that, upon entry, occasionally delivers the particle to the target and ends the trial; b is the per-frame trigger rate. Releasing the particle at the gate ($\xi = \theta$, the convention under which $b_c(\theta)$ is quoted), recording first-arrival times over many trials, and scanning b at fixed θ —best near $\theta \simeq 0.38$, where b_c is smallest and the effect largest—should reveal the second peak for $b < b_c(\theta)$ and its annihilation at b_c , with the fold collapse (14) as the sharp signature. A simulated experiment (Brownian dynamics with counting noise) confirms measurability: the robust second-peak observable $R = \Phi(\tau_p)/\Phi(\tau_v)$ (diffusive-peak over valley density) decreases through unity across the fold, and resolving b_c to ± 0.1 requires $\sim 4 \times 10^4$ delivery-and-terminate trials (App. E). We caution that standard resetting-to-origin is *not* this problem: it reinjects probability into the survival domain, whereas the correct protocol is delivery-and-terminate. (ii) *Engineered or small-world transport networks* with a one-way express edge to an absorbing node realize $-b|u\rangle\langle u|$ directly (microfluidic maze with a one-way valve; programmable packet-routing graph; electrical random-walk analog with a tunable shunt to ground; lossy photonic link); two such edges provide the cleanest test of the rank-two triple peak and cusp. (iii) *Intracellular motor capture* to a reaction site realizes it in a coarse-grained, ideally reconstituted, in-vitro setting. In each case the caveat is that only a genuinely one-way, terminating channel maps to the rank-one killing defect.

C. Non-Hermitian character

We stress what the non-Hermitian character is and is not. Q_λ is a passive open-Markov loss defect, analogous to an effective non-Hermitian Hamiltonian with a localized absorber (imaginary potential) [12]; it is *not* PT-symmetric, since there is no balanced gain and loss. The survival-sector operator is itself self-adjoint in the one-defect model (the directedness enters only the absorption vector), so “non-Hermitian” here denotes open-system probability loss, not a spectral mechanism. Crucially, b_c is *not* an exceptional point: an EP requires eigenvalue coalescence with a defective (Jordan) block, whereas we find numerically ($N = 200$) that the eigenvalues remain real, simple, and smooth through b_c , with consecutive

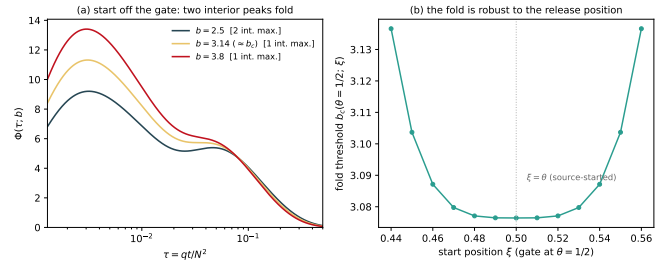


FIG. 5. Start dependence (Sec. IV E). (a) Continuum FPT density $\Phi(\tau; b)$ for a walker released off the gate ($\xi = 0.44$, $\theta = \frac{1}{2}$): both the capture and the diffusive-return features are genuine interior peaks, which annihilate through the fold as b crosses $b_c \simeq 3.14$ (two interior maxima $\rightarrow$ one). (b) The fold threshold $b_c(\theta = \frac{1}{2}; \xi)$ varies smoothly and symmetrically about the source-started point $\xi = \theta$, so the saddle-node is robust to the release position.

gaps bounded away from zero, and the fold arises from cancellation among at least three signed residues, not from spectral degeneracy. The $(b_c - b)^{1/2}$ scaling in (14) is a saddle-node of the scalar $\Phi(\tau; b)$, not EP eigenvalue splitting—the same exponent for a different object. In the reversible diffusive limit the δ -sink operator is self-adjoint (Sturm-Liouville), so no spectral EP exists at all.

VI. DISCUSSION

We have shown that shortcut-induced FPT bimodality, long known phenomenologically, is the observable signature of an exactly solvable saddle-node fold: a rank-one directed defect adds a finite-time diffusive-return peak whose birth and death, for the source-started convention, occupy a single numerically continued interval $0 < b < b_c(\theta; \xi)$, with generic $(1/2, 3/2)$ scaling, a minimal count of three modes, a rank-two extension in which a cusp organizes a triple-peak region, and the same fold morphology on finite two-dimensional lattices. A direct Monte-Carlo, an exact channel decomposition, and the start-position dependence confirm the mechanism physically. The single most defensible novelty is the bifurcation-theoretic classification with a predictive, computable threshold, distinct from the known bimodality and from the underlying defect-resolvent method. The generic fold exponents and the two-shortcut triple peak are directly measurable, on feedback-controlled colloidal search and engineered transport networks.

Beyond the results verified here, the analysis leaves a small set of open questions. A fully rigorous $N \rightarrow \infty$ statement requires, in addition to the standard convergence of fixed-mode residues (App. B), uniform C^1 convergence of the moment sums $S_{1,2,3}$ near the fold; we state the precise hypotheses and support them by the observed $O(N^{-2})$ convergence, but do not prove uniform C^1 control of the tail. The endpoint constant $B^* \approx 0.78903$

TABLE I. Independent numerical verifications (this work).

Quantity	Agreement
δ -sink jump $= D_\theta$ [Eq. (6)]	1.8×10^{-14}
I, J vs. quadrature [Eqs. (B4),(B5)]	$\sim 10^{-9}$
Identity $J = \sin(k)D_k/4b$ [Eq. (B6)]	2.9×10^{-15}
Master curve vs. exact residues ($N = 1200$)	$\lesssim 6 \times 10^{-5}$
$b_c(\theta)$ vs. indep. numerics ($\theta \in [0.2, 0.5]$)	$\sim 10^{-6}$
Endpoint $b_c \cdot \theta \rightarrow B^*$ (indep., $\theta \in [0.2, 0.27]$)	0.789026
Rank-2 secular $D_{12}(k_j)$ ($N = 800$)	$\sim 10^{-5}$
Cusp residuals $\tau^n S_n / \Phi$ ($n \leq 3$)	$\lesssim 10^{-9}$
MC π_{sc} vs. exact (4×10^5 walkers)	0.6×10^{-3}

is resolved numerically (independently, to six digits) but lacks a closed form, the large- L renormalized scaling of b_c in two dimensions is open, and whether B^* and the antipodal b_c admit elementary closed forms is unknown. None of these affects the exact finite-system results.

ACKNOWLEDGMENTS

We thank colleagues at the Bristol Centre for Complexity Sciences for discussions.

Appendix A: Exact finite- N determinant and residues

We derive Eqs. (2)–(3). Let P_0 be the $(N-1) \times (N-1)$ transient (sub-stochastic) transition matrix of the lazy ring with $v = 0$ deleted; on the interior sites it is a symmetric tridiagonal path with diagonal $1 - q$ and off-diagonal $q/2$. The directed shortcut diverts $\lambda = \beta(1 - q)$ from the self-loop at u into the absorbing target, i.e. it subtracts λ from the (u, u) entry: $P_\lambda = P_0 - \lambda |u\rangle\langle u|$, the rank-one defect of (1). The first-passage generating function to v from a start r , $\tilde{F}_r(z) = \sum_{t \geq 1} F(t)z^t$, is a ratio of resolvent elements of P_λ ; by the matrix-determinant lemma its poles are the zeros of

$$\det(I - zP_\lambda) = \det(I - zP_0) [1 + z\lambda G_{uu}^0(z)], \quad (\text{A1})$$

with $G^0(z) = (I - zP_0)^{-1}$. Introducing the spectral variable $y = 1 + (z^{-1} - 1)/q$ maps the lazy path onto the standard second-difference operator: $\det(yI - \frac{1}{2}A_{\text{path}}) = 2^{-(N-1)}U_{N-1}(y)$ and, by the cofactor structure of a tridiagonal inverse,

$$G_{uu}^0 \propto [(yI - \frac{1}{2}A_{\text{path}})^{-1}]_{uu} = \frac{2U_{u-1}(y)U_{N-u-1}(y)}{U_{N-1}(y)}. \quad (\text{A2})$$

Substituting (A2) into (A1) and clearing U_{N-1} gives the characteristic determinant $D_u(y) = aU_{N-1}(y) + 2U_{u-1}(y)U_{N-u-1}(y)$ of (2), with $a = q/\lambda$. The residue of $\tilde{F}_r$ at each root y_j produces the geometric sum (3) with

$s_j = 1 - q + qy_j$ and $B_j = q\mathcal{N}_{r,u}(y_j)/D'_u(y_j)$. Writing $A_m := U_{m-1}(y)$ (with the convention $A_0 = U_{-1} = 0$), the piecewise numerator is

$$\mathcal{N}_{r,u} = a[A_r + A_{N-r}] + 2A_{N-u}[A_r + A_{u-r}], \quad r \leq u, \quad (\text{A3})$$

and its mirror ($r \leftrightarrow N - r$, $u \leftrightarrow N - u$) for $r \geq u$. Finally, by Sherman–Morrison the eventual channel (splitting) probability—the fraction of trajectories absorbed through the shortcut—is

$$\pi_{sc}(r) = \frac{2 \min(r, u) [N - \max(r, u)]}{aN + 2u(N - u)}, \quad (\text{A4})$$

which for the antipodal case reduces to $\pi_{sc} = \rho/(a + L)$ with $\rho = \min(r, N - r)$ the start's short-arc distance to the absorbing target. We emphasize that π_{sc} is the eventual splitting probability, *not* the mass under the first peak; conflating the two is a common error we deliberately avoid. The two absorption channels of Sec. IV E have the explicit generating functions

$$\begin{aligned} \tilde{F}_r^{sc}(z) &= z\lambda \langle u | (I - zP_\lambda)^{-1} | r \rangle, \\ \tilde{F}_r^{\text{diff}}(z) &= z^{\frac{q}{2}} (\langle 1 | + \langle N - 1 |) (I - zP_\lambda)^{-1} | r \rangle, \end{aligned} \quad (\text{A5})$$

with $\tilde{F}_r = \tilde{F}_r^{sc} + \tilde{F}_r^{\text{diff}}$ [Eq. (4)]. Their t -domain inverses $f_{sc}(t) = \lambda \langle u | P_\lambda^{t-1} | r \rangle$ and $f_{\text{diff}}(t) = \frac{q}{2}(\langle 1 | + \langle N - 1 |) P_\lambda^{t-1} | r \rangle$ are the channel densities of Fig. 3(b).

Appendix B: Continuum amplitude from the defect resolvent

We derive Eqs. (8)–(9) from the resolvent of $H_b = -\frac{1}{2}\partial_x^2 + b\delta_\theta$ on $(0, 1)$ with Dirichlet ends, continued to $z = -k^2/2$ (so $\mu = k^2/2 = 2w^2$).

1. *Free resolvent.* With $\varkappa = \sqrt{2z}$ (a spectral variable, not the laziness q), $R_0(x, y; z) = 2 \sinh(\varkappa x_-) \sinh(\varkappa(1 - x_-)) / (\varkappa \sinh \varkappa)$; continued to $\varkappa = ik$,

$$r_0(x, y; k) = \frac{2 \sin(kx_-) \sin(k(1 - x_-))}{k \sin k}. \quad (\text{B1})$$

2. *Defect resolvent (Sherman–Morrison, rank-one δ -sink).*

$$r_b = r_0 - \frac{b r_0(x, \theta) r_0(\theta, y)}{1 + b r_0(\theta, \theta)} = r_0 - \frac{4b \phi(x) \phi(y)}{k \sin k D_\theta}, \quad (\text{B2})$$

with $\phi = \phi_{k,\theta}$ of (9); the pole condition $1 + b r_0(\theta, \theta) = 0$ is exactly $D_\theta(k; b) = 0$ of (6). 3. *Eigenfunction.* The residue eigenfunction is $\psi_j = \phi_{k_j, \theta}$: it is continuous at θ , satisfies the Dirichlet conditions, and obeys the δ -sink jump $\psi'(\theta^+) - \psi'(\theta^-) = 2b\psi(\theta)$, which is algebraically equivalent to $D_\theta = 0$ (verified to 1.8×10^{-14}). 4. *Residue = FPT amplitude.* The survival transform is $\bar{Q}_\xi(z) = \int_0^1 r_b(y, \xi; z) dy$ and $\bar{f} = 1 - z\bar{Q}$; the residue at $z = -\mu_j$

(using $k - k_j = -(z + \mu_j)/k_j$) gives the normalization-independent amplitude

$$G_j = \mu_j \psi_j(\xi) \frac{\int_0^1 \psi_j}{\int_0^1 \psi_j^2} = \mu_j \frac{\phi_{k_j, \theta}(\xi) I}{J}, \quad (\text{B3})$$

i.e. Eq. (8) with $\mu_j = 2w_j^2$. 5. *Closed forms.* With $A = \sin k\theta$, $B = \sin k(1 - \theta)$,

$$I(k, \theta) = \frac{1}{k} [B(1 - \cos k\theta) + A(1 - \cos k(1 - \theta))], \quad (\text{B4})$$

$$J(k, \theta) = \frac{1}{2} [\theta B^2 + (1 - \theta) A^2] - \frac{AB \sin k}{2k}, \quad (\text{B5})$$

matching direct quadrature to $\sim 10^{-9}$. 6. *Normalization identity.* At a root of (6),

$$J(k_j, \theta) = \frac{\sin(k_j) D_k(k_j; b)}{4b}, \quad (\text{B6})$$

with $D_k = \sin k + k \cos k + 2b[\theta \cos(k\theta) \sin(k(1 - \theta)) + (1 - \theta) \sin(k\theta) \cos(k(1 - \theta))]$; this exact algebraic identity (verified to 2.9×10^{-15}) ties the residue norm to the determinant derivative, and $D_k(k_j) \neq 0$ certifies each affected mode as nondegenerate. 7. *Antipodal reduction.* For $\theta = \frac{1}{2}$, $G_{\xi, 1/2} = 4w(1 - \cos w) \sin(2w\xi^*) / [\sin^2 w(1 + b(b + 2)/4w^2)]$ with $\xi^* = \min(\xi, 1 - \xi)$, recovering the previously verified antipodal amplitude. 8. *Node modes.* When $\sin(n\pi\theta) = 0$ the mode decouples from the sink and is the free Dirichlet mode $\sin(n\pi x)$, with amplitude $G_n^{\text{node}} = n\pi[1 - (-1)^n] \sin(n\pi\xi)$ (even n vanish, and all node modes vanish at $\xi = \theta$). The reconstructed master curve (affected plus node) matches exact finite- N residues to $\lesssim 6 \times 10^{-5}$ (worst case in the early- τ tail; per-mode $\sim 2 \times 10^{-4}$) with $O(N^{-2})$ convergence.

The $N \rightarrow \infty$ limit. Convergence of fixed-mode residues $G_{N,j} \rightarrow G_j$ follows from norm-resolvent convergence of the killed generator under standard hypotheses (A) $\xi_N \rightarrow \xi$, $\theta_N \rightarrow \theta$; (B) bulk convergence $H_N \rightarrow -\frac{1}{2}\partial_x^2$; (C) δ -sink form scaling $V_{N,\theta} \sim bN$; (D) a symmetrizable transient generator; (E) simple isolated poles. Passing the fold (11) to the limit additionally requires (F) uniform C^1 convergence of $S_{1,2,3}$ (including ∂_b) on a neighborhood $\tau \in [\tau_c/2, 3\tau_c/2]$ of the fold—not down to $\tau = 0$, where the heat-kernel short-time singularity lives—for which a sufficient practical bound is $\mu_{N,j} \geq cj^2$, $|G_{N,j}| \leq Cj^r$ uniformly. Under (F), fold persistence is a consequence of the implicit-function theorem: the Jacobian $\det DF_c = +S_{1,b}S_3$ is nonzero by nondegeneracy, giving a unique nearby finite- N fold with $b_{c,N} - b_c = O(\varepsilon_N)$, consistent with the observed $O(N^{-2})$. We state (F) as a hypothesis, supported by the numerics, rather than a theorem.

Appendix C: Fold criterion: proofs and asymptotics

Fold criterion. From $\Phi = \sum_j G_j e^{-\mu_j \tau}$, $\Phi' = -\sum_j G_j \mu_j e^{-\mu_j \tau} = -S_1$ and $\Phi'' = \sum_j G_j \mu_j^2 e^{-\mu_j \tau} = S_2$,

so an interior extremum is $S_1 = 0$ and two extrema merge at $S_1 = S_2 = 0$, Eq. (11). The parameter derivative of a root of (6) is $\partial_b k_j = -D_b/D_k$ with $D_b = 2 \sin(k\theta) \sin(k(1 - \theta))$ and D_k as in (B6), which propagates into $\partial_b S_1$ and gives the nondegeneracy condition needed for a structurally stable fold.

Minimal-mode theorem. As in Sec. IIID, $S_1 = S_2 = 0$ has no two-mode solution with $A_j \neq 0$; the minimal count is three, with the alternating-sign ratio $A_1 : A_2 : A_3 = (\mu_3 - \mu_2) : (\mu_1 - \mu_3) : (\mu_2 - \mu_1)$. This retires an earlier (incorrect) two-mode heuristic and shows the fold is intrinsically a many-mode cancellation.

Endpoint law. As $d = \min(\theta, 1 - \theta) \rightarrow 0$ the gate approaches the absorbing boundary. Rescaling $y = x/d$ and $B = bd$ and taking $d \rightarrow 0$ gives a half-line boundary-layer problem: diffusion on $(0, \infty)$ with an absorbing end at 0 and a δ -sink of strength B at $y = 1$, started at $y = 1$. Its spectrum is fixed by a transcendental condition of the same $\sinh/\sin$ form as (6), and imposing the fold $S_1 = S_2 = 0$ in this limit selects a single strength B^* and reduced time, whence $b_c(\theta) \sim B^*/d$ and $\tau_c \sim c_* d^2$ [Eq. (12)]. Resolving $b_c(\theta)$ independently by bisection on the fold count—with the root search extended to the high- k fold mode ($w \sim 1.78/d$), which the naive locator misses—gives $b_c(\theta)\theta = 0.789026$ stably for $\theta \in [0.20, 0.27]$ and a linear extrapolation $B^* \approx 0.78903$ (with $c_* \approx 0.158$); this reproduces the constant to six digits without any reference table, refuting a circular check. Below $\theta \simeq 0.15$ the fold mode is too high- k for clean resolution, and a closed form for B^* remains open. Near the antipode a regular perturbation of $\tan w = -2w/b$ yields $b_c(\frac{1}{2} + \epsilon) \approx 3.0764 - 133\epsilon^2$, with the antipodal values (13) from $S_1 = S_2 = 0$ directly ($S_3 \approx 2.8 \times 10^5$, $\partial_b S_1 \approx 22$, nondegenerate). The truncated thresholds ($K = 2$ none, $K = 3 \rightarrow 3.046$, $K = 4$ none, $K = 5 \rightarrow 3.076$) confirm rapid, parity-structured convergence.

Appendix D: Two-shortcut determinant and cusp

For two sources the secular condition is $\det[I_2 + BW(k)] = 0$ with $W_{ab} = r_0(k; \theta_a, \theta_b)$ the interval resolvent (B1); expanding the 2×2 determinant gives $D_{12}(k)$ of (16), whose cross term $\propto b_1 b_2$ is the genuine two-defect interaction (it $\rightarrow 0$ as $\alpha \rightarrow \eta$, recovering a single defect of strength $b_1 + b_2$). The rank-two residue eigenfunction is $\psi_j(x) = -\sum_a b_a c_a^{(j)} r_0(k_j; x, \theta_a)$ with $c^{(j)} \in \ker[I_2 + W(k_j)B]$, and $G_j = \mu_j \psi_j(\xi) \int \psi_j / \int \psi_j^2$ with $\int \psi_j = -\sum_a b_a c_a I_{\theta_a}$ and $\int \psi_j^2 = \sum_{ab} b_a c_a b_b c_b C_{ab}$; the constituents are I_θ [Eq. (B4)] and $C_{ab} = \int_0^1 r_0(x, \theta_a) r_0(x, \theta_b) dx$, both elementary. The cusp solves $S_1 = S_2 = S_3 = 0$ with $S_4 \neq 0$ and a nonzero unfolding Jacobian $\det \partial(S_1, S_2)/\partial(b_1, b_2)$. At the located point (17) an independent reimplementatation gives normalized residuals $\tau S_1/\Phi \approx 7 \times 10^{-12}$, $\tau^2 S_2/\Phi \approx 5 \times 10^{-11}$, $\tau^3 S_3/\Phi \approx 4 \times 10^{-10}$, $\tau^4 S_4/\Phi \approx -1.67$, and $\det \partial(S_1, S_2)/\partial(b_1, b_2) \approx -6.9 \times 10^7$, confirm-

ing a nondegenerate codimension-two cusp.

Appendix E: Numerical methods and reproducibility

Exact finite- N FPT poles are computed from the symmetric transient block via `numpy eigvalsh`, with amplitudes from the closed-form residues; high-precision cross-checks use `mpmath`. Continuum spectra are the roots of (6)/(16), and amplitudes use (8) and App. D. All scripts are deterministic. Table I collects the verification tolerances quoted throughout. The figures are generated by `dpma_prr_figures.py` (results) and `dpma_schematic.py` (model); the existence boundary by `dpma_saddle_node_bc_theta.py`; the amplitude/master curve by `dpma_general_u_master_curve.py`; the fold scaling by `dpma_saddle_node_certification.py`; the cusp by `dpma_cusp_verify.py`; the rank-two determinant and triple peak by `dpma_multishortcut.py`; the 2D test by `dpma_2d_universality.py`; the direct-simulation, channel-resolution, and non-antipodal panels by `dpma_channel_mc.py`; the start-dependence panels by `dpma_start_dependence.py`; the independent Brownian-dynamics instance by `dpma_brownian_fold.py`; the endpoint-constant resolution by `dpma_endpoint_law.py`; and the measurability estimate by `dpma_measurement.py`. The 1D boundary $b_c(\theta)$ is obtained by bisection on the valley-peak-pair count over $\theta \in [0.2, 0.5]$, $b \in [0.5, 20]$, retaining the affected roots with $k \leq 40$; over this range the two-mode set is a single connected interval in b (no re-entrant window). The 2D fold is computed at $L = 31$ by extremum merging (the second interior maximum's prominence $\rightarrow 0$ near $\beta \approx 0.69$), not the ad-hoc valley heuristic. The Monte-Carlo uses 4×10^5 vectorized walkers with a fixed seed, logarithmic τ -binning, and shortcut-vs-diffusive absorption events recorded per walker; the quoted π_{sc} uncertainty is the binomial standard error. *Code availability.* All scripts are deterministic and will be released with the paper (repository archive/commit hash to be added).

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